Tianchu Hang

tiahang@bu.edu/ +1 (469) 980-0505 /LinkedIn
Evanston, IL, 60201

EDUCATION BACKGROUND

Boston University, Boston **Jun 2021 - May 2022**
Degree Conferred: Master of Science
Major: Population Health Research – Data Science
Scholarship honored from School of Public Health

University of California, Berkeley **Aug 2018 - Aug 2020**
Degree Conferred: Bachelor of Arts
Major: Data Science

Peralta Community College District **Aug 2016 - May 2018**
Major: Statistics (Credits earned transferred to UCB)
Academic Honor Student, Fall 2016 & Spring 2017 & Fall 2017

PUBLICATIONS

1. **Tianchu Hang**, William H Moore, Samrachana Adhikari, Yu Chen, & Stella K Kang (2025). Predicting Inappropriate Follow-Up Care for Incidental Pulmonary Nodules Using Machine Learning Models. – Manuscript in preparation.
2. Wenhan Lv, Zhongquan Jian, Nianxin Ke, **Tianchu Hang**, Qingqi Hong, Qingqiang Wu (2025). Multimodal Fusion Diffusion Modeling: An Innovative Framework for 3D Human Action Generation. ICCV 2025 Conference. – Manuscript under review.
3. Nitya L Thakore, Rienna Russo, **Tianchu Hang,** William H Moore, Yu Chen, & Stella K Kang (2023). Evaluation of Socioeconomic Disparities in Follow-Up Completion for Incidental Pulmonary Nodules. *Journal of the American College of Radiology*, *20*(12), 1215-1224.
4. Fang Zhang, **Tianchu Hang**, & Yang Bai (2022, March). Hybrid Weibo Tags and Topic Mining for User Similarity Model. In CIBDA 2022; 3rd International Conference on Computer Information and Big Data Applications (pp. 1-4). VDE.
5. **Tianchu Hang**, Yang Bai, & Guishi Deng (2020). A Method for Community Partition Based on Information Granularity. In Security and Privacy in Social Networks and Big Data: 6th International Symposium, SocialSec 2020, Tianjin, China, September 26–27, 2020, Proceedings 6 (pp. 407-419). Springer Singapore.

RESEARCH EXPERIENCES

Research Engineer, NYU Langone Health (full-time) **Sep 2022 – Sep2023**
Supervisor: Professor Stella Kang, NYU School of Medicine (currently at Columbia University)

- Developed and validated a machine learning model in Python, achieving 88% AUC for predicting inappropriate care of patients with incidental lung nodules (ILNs) using electronic health records (EHR) and social determinants of health data.
- Conducted multivariable regression analysis on EHR data from a large urban tertiary center to examine the association between social vulnerability, residential segregation, and follow-up disparities for ILNs, informing policy and intervention strategies.
- Designed and assisted in decision analysis for ILN intervention, projecting tobacco cessation and community-based care navigation bundles to reduce disparities in lung cancer detection and mortality, identifying tailored strategies for diverse communities.
- Built and optimized ETL pipelines for 100GB+ of EHR data using SQL and R, resolving data inconsistencies and handling imbalanced/missing datasets with grid search optimization and multivariate imputation methods in R.
- Worked collaboratively with cross-functional teams to establish and manage a radiology data hub within an AWS data lake, integrating 15GB+ of healthcare data spanning patient-level to neighborhood-level for streamlined radiology research analysis.
- Contributed to grant proposal writing and data visualization, effectively communicating research insights for funding and publication.

Graduate Researcher, Uncounted Lab, Boston University (12 h/w) **Nov 2021 - May 2022**
Supervisor: Professor Andrew C. Stokes, Boston University

- Cleaned and analyzed biomarker and demographic data from 45,000 participants (Population Assessment of Tobacco and Health study) to evaluate cardiovascular risk associated with e-cigarette use, using Stata and R.
- Built and applied unsupervised clustering models (K-means, PCA) to identify phenotypic clusters related to tobacco use, adjusting for skewed data distributions.
- Standardizing biomarker and demographic data, addressing missing values, and correcting skewed distributions through log transformations.
- Prepared visualizations and examined associations between cluster membership, cardiovascular risk, and e-cigarette usage patterns.

Undergraduate Researcher, UC Berkeley (10 h/w) **Aug 2020 - Nov 2020**
Project: Analyzing Public Sentiment and Policy Impact in Germany's COVID-19 Response
Supervisor: Professor Deniz Gokturk, UC Berkeley

- Utilized Python to conduct web scraping through the unofficial Twitter API, extracting posting time, content, and geolocation details for hashtags like #FlattenTheCurve, #SocialDistancing, and #StayHome, capturing public sentiment during Germany's COVID-19 second wave.
- Employed Matplotlib to create visualizations comparing regional variations in adherence to public health measures across German states, highlighting differences in mask compliance and social distancing behaviors.
- Conducted sentiment analysis on #CoronaWarnApp to evaluate public reception of Germany's COVID-19 contact tracing app, uncovering trends in adoption and resistance across demographic groups.
- Performed time series analysis to map shifts in public sentiment toward #SocialDistancing, providing insights into the effectiveness of public health messaging during the pandemic.

Undergraduate Researcher, Dalian University of Technology (10 h/w) **Sep 2019 - Aug 2020**
Supervisor: Professor Guishi Deng, Dalian University of Technology

- Developed PWRALP (Product-weighted RALP) to analyze user interactions in social networks and uncover hidden social relationships, achieving 88% accuracy, outperforming CN, WRA, and WRALP.
- Performed data cleaning and preprocessing on social network datasets to extract effective experimental data for community partitioning analysis.
- Conducted algorithm experiments to predict user similarity values and evaluated AUC and Precision indices to measure model accuracy.

PROFESSIONAL EXPERIENCES

Data Scientist, North Carolina Department of Health and Human Services (full-time) **Mar 2024 - Now**

- Designed quantitative methods, including χ^2, t-tests, ANOVA, and comparative interrupted time series (CITS) models, to evaluate the impact of mental health service investments.
- Developed and maintained ETL pipelines to extract, integrate, and preprocess 100GB+ of healthcare claims data from IBM Cloud using SQL, Python, and Tableau Prep, ensuring data accuracy and consistency.
- Conducted statistical analyses to identify geographic disparities, service overlap rates, and readmission trends, informing resource allocation and evidence-based policy decisions.
- Built automated Tableau dashboards to provide real-time insights into behavioral health crisis services utilization, supporting the Division's five-year Strategic Plan.

Graduate Student Instructor, Boston University (10 h/w) **Jan 2022 - May 2022**

- Taught an introductory R lab to 30+ students and create new versions of lab materials to deliver lecture material Held weekly office hours where students may get help on programming homework, labs, and course concepts.
- Collaborated with the professor weekly; Students' grades were boosted by 18.7% from last semester.

Data Analyst, Zach Technology (full-time) **Nov 2020 – June 2021**

- Developed Tableau dashboards to monitor KPIs for hosts and channel performance metrics, and conducted reporting across departments to analyze program topics, resulting in a 20% increase in audience engagement.
- Conducted in-depth statistical data analysis of 7GB of advertisement data on our website using Google Cloud Platform (BigQuery SQL and Looker) to automate dashboards and presented insights on market trends and pricing strategies to clients.
- Integrated and preprocessed the email status tracking data and generated data visualization with Matplotlib, used the A/B testing, and supported all external communications that eventually boosted the installed audience by 15.7%.
- Developed Python web crawlers to scrape 100+ YouTube channels to generate detailed datasets on channel analytics.

PROJECTS INVOLVEMENTS

Alzheimer's Disease Prediction from MRI **Jan 2022 - May 2022**

- Built a regression model using feature engineering and feature selection method in MATLAB to predict persons who have Alzheimer's disease from those who have normal cognition by reading pixel values from 2D MRI slices.
- Applied Fitted Linear Regression and cross-validation method and reached 85.7% accuracy on the test set.

Pac-Man **Jan 2021 - March 2021**

- Developed Python Pac-Man projects employing informed state-space search, probabilistic inference, and reinforcement learning.
- Implemented various search algorithms (depth-first search, breadth-first search, A* search) to guide Pac-Man in finding food while evading ghosts.
- Designed Pac-Man agents utilizing minimax, alpha-beta pruning, expectimax search, and Q-learning techniques for a simulated robot controller.
- Utilized probabilistic inference methods (variable elimination, exact inference, joint particle filtering) to create Pac-Man agents equipped with sensors for detecting and consuming invisible ghosts, alongside implementing diverse search algorithms and heuristics for optimizing pathfinding.

Uber Passenger Tip Prediction **June 2019 - July 2019**

- Built classification models and performed Exploratory Data Analysis (EDA) to understand the dataset's features, distributions, and correlations, and for data cleaning and feature engineering.
- Applied Feature Selection filter method on data by Chi-square, F-score, and mutual information.
- Fitted Logistic Regression, Stacking, Ensemble. Fine-tune Random Forest models using GridSearch Cross Validation.

PRESENTATIONS

1. "Cluster Analysis of Cardiopulmonary Risks Associated with E-cigarette Use", Boston University, Population Health Research Symposium, May 2022
2. "A Method for Community Partition Based on Information Granularity", Security and Privacy in Social Networks and Big Data: 6th International Symposium, Aug 2020

ACTIVITIES

Core Member, Chinese Student Scholar Association, UC Berkeley	**Aug 2018 - May 2019**
Vice President, Tzu Ching Club, Berkeley City College	**Aug 2017 – Dec 2018**
Volunteer, Yanglin Elementary School	**June 2018 - Aug 2018**
Senator, Associated Students of Berkeley City College	**Aug 2017 - May 2018**
Volunteer, Asia Resources Center	**Nov 2017 – March 2018**
Volunteer, North Berkeley Senior Center	**June 2017 – Jan 2018**
Vice President, AMATYC Math Competition Club, Laney College	**March 2017 – Dec 2017**
Volunteer, American Red Cross	**Aug 2017 – Dec 2017**
Math Tutor and TA, Laney College	**Jan 2017 – June 2017**

SKILLS

Skills: Machine Learning, Predictive Analysis, Data Visualization, A/B Test, Data Cleaning, Regression Analysis, Feature Engineering, Decision Analysis, Simulation, Optimization, Calibration, Data Imputation, Data Management
Languages & Tools: Python, R, SQL, Tableau, Tableau Prep, SAS, MATLAB, Google Cloud, AWS, ArcGIS
Frameworks & Libraries: PyTorch, Scikit-learn, Pandas, NumPy, SciPy, Matplotlib, Dplyr, Ggplot2, Tidyr, Mlr3